

Tutorial 2. Camera Models, Filters, Events and Edges

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Camera models

Learning Objectives:

- Understand fundamental principles of camera optics and lens systems
- Analyze event-based camera operation and constant brightness assumption
- Master composite optical system modeling using ray tracing matrices
- Apply geometric optics principles to real-world camera configurations

Key Concepts: Thin lens formula, field of view, geometric optics, ray tracing, event-based cameras.

1.

Thin lens formula 1: A camera is equipped with a lens with focal length $f = 50$ mm and aperture diameter $A = 2$ cm. An object at 5 m from the camera is in focus. What is the minimum thickness of the camera? Is the aperture diameter needed for this calculation?

2.

Thin lens formula 2: A camera is equipped with a lens with focal length $f = 40$ mm and aperture diameter $A = 2$ cm. The distance between the image plane and the lens plane is 5 cm. What is the blur diameter of an object at infinity?

3.

Field of view: A simple camera is made of a CCD sensor and a single lens. The size of a CCD sensor is $4 \text{ cm} \times 4 \text{ cm}$. The focal length of the lens is $f = 50$ mm. What is the field of view (FOV) of this camera?

4.

Geometric optics: A simpler way to handle lenses and to determine the imaging equations is to use a *ray representation*. Instead of considering the 3D point in space, we consider a ray in space (a line) and how this is deflected by a lens when it falls within its aperture.

Let us consider a lens with focal length f and a line incident at an angle θ with the lens (the angle is with respect to the normal to the lens plane) at a (signed) height d from the optical center. Compute the exiting angle ϕ by using the two basic rules of a thin lens:

- (a) rays through the optical center are not deflected;
- (b) objects at infinity are focused at a distance f behind the lens (this means that parallel rays converge at a point at a distance f behind the lens)

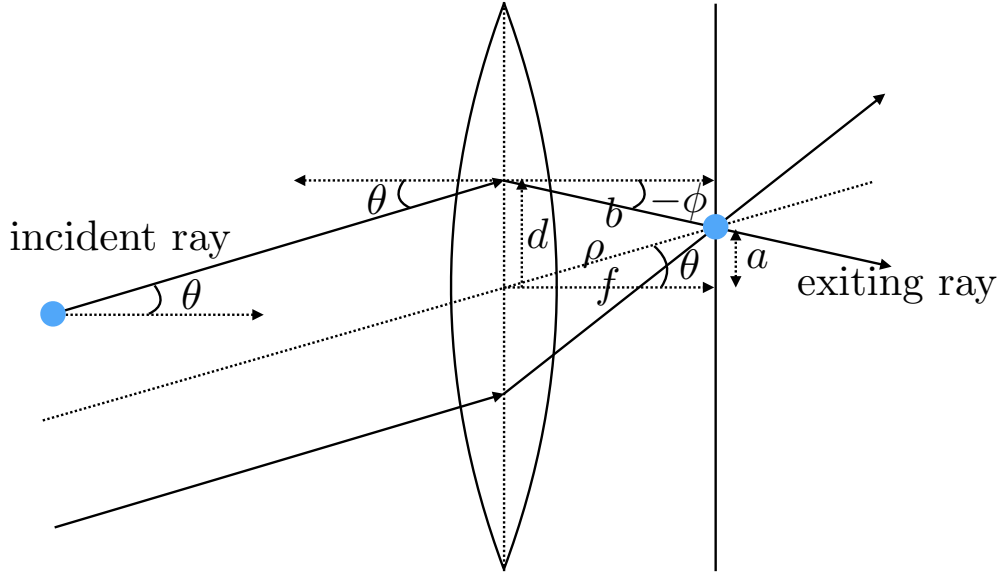


Figure 1. Geometric optics.

See Fig. 1.

Composite optical systems (challenging): Let us consider a camera with 2 lenses and let us use the geometric optics approximation to determine the imaging equations.

5. Suppose that the first lens is centered at the optical axis and has focal length f_1 . The second lens is behind the first lens at a distance Δ_1 , has focal length f_2 and its center is displaced at a height b from the optical axis.

For simplicity we assume that both lenses have infinite aperture. The camera sensor is behind the second lens at a distance Δ_2 . A point P is in front of the lens and emits light rays (see Fig. 2).

Use the optical axis as the z -axis, place the first lens at $z = 0$, and let the object coordinate satisfy $P_Z < 0$. Thus, the propagation distance from P to the first lens is $-P_Z$. The distances Δ_1 and Δ_2 are the physical free-space separations drawn in the figure. Determine the ray tracing matrix M that propagates a ray state from P to the sensor. If the ray state is instead initialized at the first-lens plane, omit the initial propagation from P .

Event-based camera: Using the constant-brightness assumption and the contrast-threshold event model, derive the event-triggering condition and explain why events are primarily caused by moving edges. State when a strong edge may nevertheless fail to trigger an event.

- 6.

Filters

Learning Objectives:

- Master various filtering techniques and their mathematical foundations
- Understand convolution operations and filter design principles
- Analyze advanced filtering methods like non-local means and mean shift

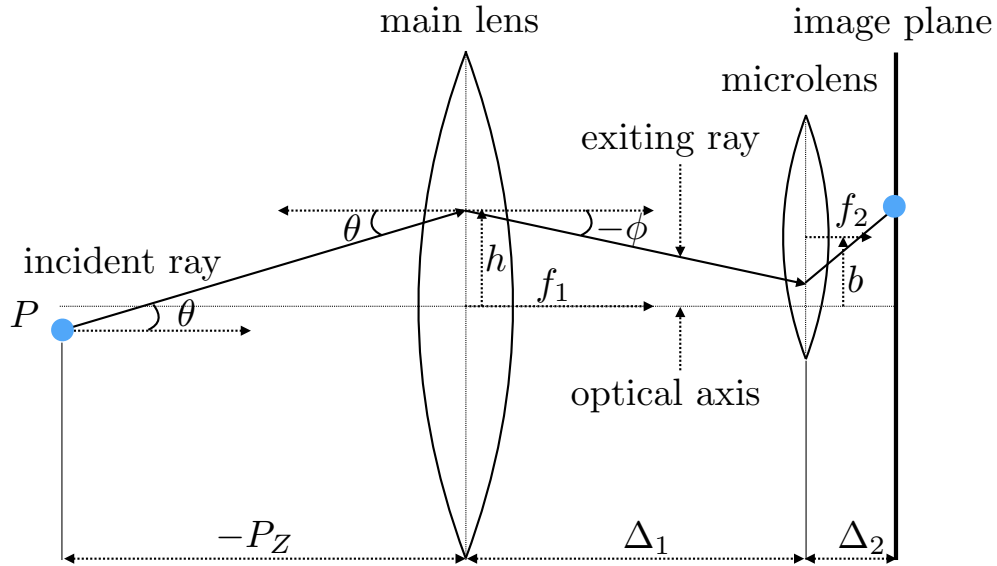


Figure 2. Geometric optics with multiple lenses.

- Apply filtering concepts to noise reduction and image enhancement

Key Concepts: Convolution, median filtering, averaging filters, non-local means, mean shift, kernel design.

Median filter [Python implementation is available¹]: The median filter takes the middle element after sorting the elements in a patch. While this operation is quite robust to non-Gaussian noise, it can lead to a severe flattening of the texture when the size of the patch is large. How could we modify the median filter so that it retains the original texture while removing outliers?

1. Consider the 1×7 image $\mathbf{I}$ below, where each pixel intensity is denoted with a letter $a, \dots, g$:

a	b	c	d	e	f	g
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- (a) Apply a 1×3 averaging filter twice to the image I , that is, apply the averaging filter to I to obtain a smoothed image J and then apply the same filter again to J . Use zero padding to calculate the convolution of edge pixels if needed. Give the expression for the center pixel of the resulting image as a function of the original pixel values in I . Show all your calculations.

- (b) Define a single 1D filter that, when applied only once to the image I , will produce the same result as applying the previous 1×3 averaging filter twice. Specify the dimension of the filter and its values.

3.

Given the image patches I and I' , find a horizontal 1D convolutional filter k that transforms the image patch I into I' . Assume the patch I is padded with zeros (just one value on each side of the patch), so the valid output size of the convolution $I * k$ matches the size of I' . Show your calculations.

$$I * k = I'$$

$$\begin{bmatrix} 128 & 128 & 128 \\ 64 & 192 & 64 \\ 0 & 128 & 64 \end{bmatrix} * k = \begin{bmatrix} 96 & 128 & 96 \\ 80 & 128 & 80 \\ 32 & 80 & 64 \end{bmatrix}$$

4.

Non-local means filter (challenging): The non-local means filter performs averaging of patches that are sufficiently similar. Under the simplifying assumption that the selected patches share the same clean patch and that their noise terms remain independent, suppose that the original image is affected by zero-mean i.i.d. Gaussian noise. What is the distribution of the denoised patch $\hat{P}$?

If we extract a patch P from the noisy image, then we select other patches P_i such that

$$\|P - P_i\|_F < d \quad (1)$$

with a certain scalar $d > 0$. If N patches P_i satisfy the constraint above, then the updated/denoised patch $\hat{P}$ is obtained via averaging

$$\hat{P} = \frac{1}{N} \sum_{i=1}^N P_i \quad (2)$$

Suppose that the original image is affected by zero-mean i.i.d. Gaussian noise. Then, we can write

$$P = P_0 + n \quad P_i = P_0 + n_i, \quad (3)$$

where P_0 is the shared noise-free patch and $n, n_i \sim \mathcal{N}(0, \sigma^2 I_d)$ are independent Gaussian random variables under this simplified model. In a practical non-local means algorithm, selecting patches according to their observed similarity can introduce statistical dependence; this exercise intentionally ignores that effect.

5.

Mean shift filter (challenging): The *mean shift* filter is a procedure to find the maxima of a probability density function (the modes) from its samples. For the Epanechnikov/hard-neighborhood setting used below, find the relation between one mean shift update and the non-local mean filter above.

The first step is to assume that the probability density function can be approximated by its kernel density estimate

$$p(x) = \frac{1}{N} \sum_{i=1}^N K\left(\frac{|x - x_i|^2}{h}\right), \quad (4)$$

where $K(|x|^2)$ is a kernel that integrates to 1 in x and h is a bandwidth parameter. A necessary condition to determine the maxima of a function is that the first order derivatives must be zero. By taking the first order derivatives in x we obtain

$$\nabla p(x) = \frac{1}{N} \sum_{i=1}^N 2 \frac{x - x_i}{h} K'\left(\frac{|x - x_i|^2}{h}\right) = 0 \quad (5)$$

Next we rearrange the equation above to obtain the following iterative update

$$x^{t+1} = \frac{\sum_{i=1}^N x_i K' \left(\frac{|x^t - x_i|^2}{h} \right)}{\sum_{i=1}^N K' \left(\frac{|x^t - x_i|^2}{h} \right)} \quad (6)$$

This update is the *mean shift* algorithm and it can be shown to converge to the modes of the distribution $p(x)$ in a number of cases.

Let us consider the Epanechnikov kernel

$$K(z) = \begin{cases} \frac{3}{4}(1-z) & 0 \leq z \leq 1 \\ 0 & z > 1 \end{cases} \quad (7)$$

then, we have

$$K'(z) = \begin{cases} -\frac{3}{4} & 0 \leq z \leq 1 \\ 0 & z > 1 \end{cases} \quad (8)$$

and the meanshift iteration becomes

$$x^{t+1} = \frac{\sum_{i=1}^N x_i \delta(|x^t - x_i|^2 < h)}{\sum_{i=1}^N \delta(|x^t - x_i|^2 < h)} \quad (9)$$

where $\delta(\text{event})$ is the indicator function of «event» (it is 1 if the event is true and 0 otherwise).

Edges

Learning Objectives:

- Understand the mathematical foundations of gradient-based edge detection
- Analyze self-similarity properties and their relationship to image gradients
- Connect geometric properties to directional derivatives and orthogonality conditions
- Apply gradient analysis to characterize image structure and features

Key Concepts: Image gradients, self-similarity, directional derivatives, orthogonality, edge characterization.

1. **Gradients 1:** Let us consider an image I . Let also $p, q \in \mathbb{R}^2$ be two pixels (represented as two 2D vectors) in I . We define *self-similarity* as the property that rays originating at p should have constant image intensity. How does the gradient $\nabla I[q]$ in the point q relate to the self-similarity in the limit as $\alpha \rightarrow 1$?

We define *self-similarity* as the property that

$$I[q] = I[p + \alpha(q - p)] \quad \forall q : \|q - p\|_2^2 \leq \rho \quad (10)$$

where $\rho > 0$ is the radius of a ball around p and $0 < \alpha < 1$. See Fig. 3.

In other words, rays originating at p should have constant image intensity. Notice that, in particular, the self-similarity property is satisfied at corners and at edges.

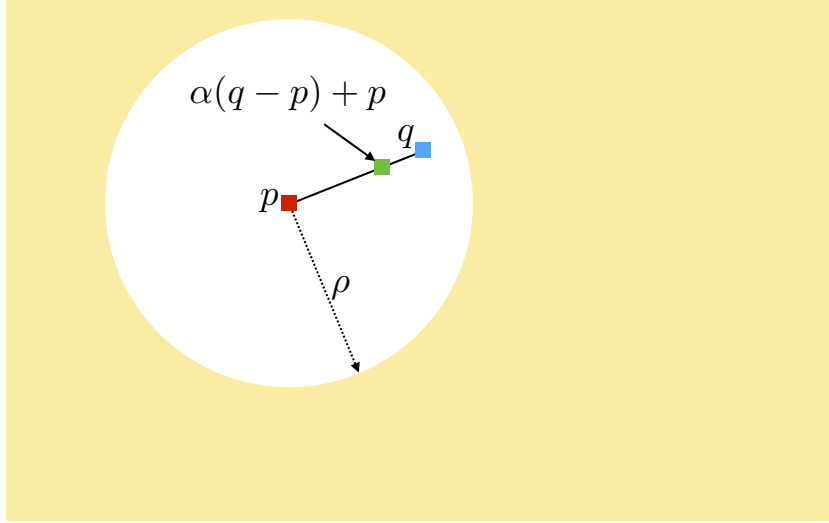


Figure 3. Gradients and self-similarity.

2. **Gradients 2:** Show that the family of all images I that can be written as a function of only the direction of $v = q - p$, with $v \neq 0$, satisfies the orthogonality property below for a fixed pixel p .

Show that the family of all images I that can be written as a function of only the direction of $v = q - p$

$$I[q] = f\left(\frac{v}{\|v\|_2}\right) = f\left(\frac{q - p}{\|q - p\|_2}\right) \quad (11)$$

for any function f , satisfies the property below for a fixed pixel p

$$\nabla I[q]^\top (q - p) = 0 \quad \forall q \neq p \quad (12)$$